CO-NORMALIZER ORTHOGONAL JOIN REDUCTION AND RATIONAL CHERN–SIMONS VOLUME ON ONE-ANGLE SPHERICAL BAR CYCLES

ARGUS OPEN RESEARCH PROJECT

Abstract. A conjecture in low-dimensional topology asks whether the SU(2) Chern–Simons invariant of every flat connection on a closed 3-manifold is rational modulo 1. For the one-angle spherical bar-cycle construction in the Bloch/scissors congruence group, we prove that the co-normalizer flatness constraint forces an orthogonal spherical join, and that vanishing of the Dehn invariant on the resulting cycle forces its normalized Chern–Simons volume to be rational.

1. Main Rationality Reduction

Theorem 1.1. *In the one-angle spherical bar-cycle family, imposing the co-normalizer holonomy constraint decomposes every admissible tetrahedron cycle as an orthogonal spherical join; tensoring the Schlafli/Dehn invariant over $\mathbb{Q}$ shows that whenever the Dehn invariant vanishes, the dihedral angles are rational multiples of π and the normalized Chern–Simons invariant lies in $\mathbb{Q}/\mathbb{Z}$.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 2862 in the domain of low-dimensional topology and algebraic K-theory follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *scoped rationality theorem* deciding the scope: co-normalizer flatness forces an orthogonal spherical join and rational normalized Chern-Simons volume whenever the Dehn invariant vanishes.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- unrestricted SU(2) Chern-Simons rationality.

References

[1] Primary problem list and source literature for Result 2862 (low-dimensional topology and algebraic K-theory), archived with exact theorem locators and verification certificates in the accompanying repository package.

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